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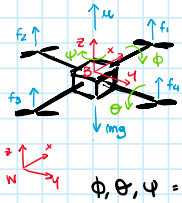


$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\phi & -s\phi \\ 0 & s\phi & c\phi \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}$$

$$R_z(\psi) = \begin{bmatrix} c\psi & -s\psi & 0 \\ s\psi & c\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\phi, \theta, \psi = f(t)$$

$$R_W = R_z(\psi) \cdot R_y(\theta) \cdot R_x(\phi) = \begin{bmatrix} c\theta c\psi & c\psi s\theta s\phi - c\phi s\psi & s\phi s\psi + c\phi c\psi s\theta \\ c\theta s\psi & c\phi c\psi + s\theta s\phi s\psi & c\phi s\theta s\psi - c\psi s\phi \\ -s\theta & c\theta s\phi & c\theta c\phi \end{bmatrix}$$

$$R_B = [R_{BW}]^T$$

$$W_B = -D[R_B, t] \cdot \text{Inverse}[R_B] = \begin{bmatrix} 0 & s\phi\theta' - c\theta c\phi\psi' & c\phi\theta' + c\theta s\phi\psi' \\ -s\phi\theta' + c\theta c\phi\psi' & 0 & -\phi' + s\theta\psi' \\ -c\phi\theta' - c\theta s\phi\psi' & \phi' - s\theta\psi' & 0 \end{bmatrix}$$

$$W = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & \omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \rightarrow \begin{aligned} \omega_x &= \phi' - s\theta\psi' \\ \omega_y &= c\phi\theta' + c\theta s\phi\psi' \\ \omega_z &= -s\phi\theta' + c\theta c\phi\omega' \end{aligned}$$

equivalente:

$$W_W = D[R_W, t] \cdot \text{Inverse}[R_W]$$

$$W_B = R_B \cdot W_W \cdot R_W$$

states : $q = \begin{Bmatrix} x \\ y \\ z \\ \psi \\ \theta \\ \phi \end{Bmatrix}$ (ξ: x, y, z; η: ψ, θ, ϕ) $\quad \dot{q} = \begin{Bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{Bmatrix} \rightarrow \{q, \dot{q}\}$

$$\Omega = W_\eta \eta' \Rightarrow \begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{Bmatrix} = \underbrace{\begin{bmatrix} -s\theta & 0 & 1 \\ c\theta s\phi & c\phi & 0 \\ c\theta c\phi & -s\phi & 0 \end{bmatrix}}_{W_\eta} \begin{Bmatrix} \psi' \\ \theta' \\ \phi' \end{Bmatrix}$$

EULER-LAGRANGE METHOD

$$\mathcal{L} = K - P$$

$$P = mgz$$

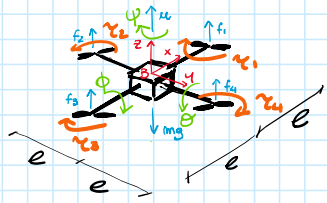
$$K = K_{trasl} + K_{rot} = \frac{1}{2}\dot{\xi}^T m \dot{\xi} + \frac{1}{2}\Omega^T I \Omega$$

$$= \frac{1}{2} m \dot{\xi}^T \dot{\xi} + \frac{1}{2}(W_\eta \dot{\eta})^T I (W_\eta \dot{\eta}')$$

$$\frac{1}{2}\dot{\eta}^T \underbrace{(W_\eta^T I W_\eta)}_{J} \dot{\eta}$$

$$I = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \qquad J = \begin{bmatrix} I_z c^2\theta c^2\phi + I_x s^2\theta + I_y c^2\theta s^2\phi & I_y c\theta c\phi s\phi - I_z c\theta c\phi s\phi & -I_x s\theta \\ I_y c\theta c\phi s\phi - I_z c\theta c\phi s\phi & I_y c^2\phi + I_z s^2\phi & 0 \\ -I_x s\theta & 0 & I_x \end{bmatrix}$$

eq. moto : $\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \frac{\partial \mathcal{L}}{\partial q} = \begin{Bmatrix} F_\xi \\ \tau \end{Bmatrix}$



$F_\xi = R \cdot \hat{F}$ where $\hat{F} = \begin{bmatrix} 0 \\ 0 \\ u \end{bmatrix}$ $\quad u = \sum_{i=1}^{4} f_i$

$$= \begin{bmatrix} u(c\phi c\psi s\theta + s\phi s\psi) \\ u(-c\psi s\phi + c\phi s\theta s\psi) \\ u c\theta c\phi \end{bmatrix} \quad f_i = k\omega_i^2$$

$$\tau = f \cdot b = \begin{Bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{Bmatrix} = \begin{bmatrix} (f_2 - f_4)e \\ (f_3 - f_1)e \\ \sum \tau_i \end{bmatrix} \Rightarrow \bar{\tau} = W_\eta^T \cdot \tau = \begin{bmatrix} -(f_2 - f_4)e s\theta + c\theta(\sum \tau_i c\phi + (f_3 - f_1)e s\phi) \\ (f_3 - f_1)e c\phi - \sum \tau_i s\phi \\ (f_2 - f_4)e \end{bmatrix}$$

$$\rightarrow \frac{d}{dt}\left(\frac{\partial L_{trans}}{\partial \dot{\xi}}\right) - \frac{\partial L_{trans}}{\partial \xi} = F_\xi \qquad m I_{3\times3} \ddot{\xi} + \begin{Bmatrix} 0 \\ 0 \\ mg \end{Bmatrix} = F_\xi$$

$$\rightarrow \frac{d}{dt}\left(\frac{\partial L_{rot}}{\partial \dot{\eta}}\right) - \frac{\partial L_{rot}}{\partial \eta} = \tau \qquad J(\eta)\ddot{\eta} + \underbrace{\cancel{C(\eta, \dot{\eta})\dot{\eta}}}_{0} = \tau$$

eq. dynamic : $M(q)\ddot{q} + g(q) = S^T(q) \cdot U$

$$\begin{bmatrix} m I_{3\times3} & 0_{3\times3} \end{bmatrix} \begin{Bmatrix} \ddot{\xi} \\ \ddot{\eta} \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ mg \\ 0 \\ 0 \\ 0 \end{Bmatrix} = \begin{Bmatrix} F_\xi \\ \tau \end{Bmatrix}$$

eq. dynamic: $M(q)\ddot{q} + g(q) = S(q)\,u$

$$\begin{bmatrix} m I_{3\times3} & 0_{3\times3} \\ 0_{3\times3} & J(\eta) \end{bmatrix} \begin{Bmatrix} \ddot{\xi} \\ \ddot{\eta} \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ mg \\ 0 \\ 0 \\ 0 \end{Bmatrix} = \begin{Bmatrix} F_\xi \\ \tau \end{Bmatrix}$$

$$u: \begin{Bmatrix} T \\ \tau_x \\ \tau_y \\ \tau_z \end{Bmatrix} = \begin{Bmatrix} \sum_i f_i = u \\ {}^1\ell(f_2 - f_4) \\ {}^4\ell(f_3 - f_1) \\ \sum_i \tau_i \end{Bmatrix}$$

$$\begin{Bmatrix} F_\xi \\ \tau \end{Bmatrix} = \begin{bmatrix} u(c\phi c\psi s\theta + s\phi s\psi) \\ u(-c\psi s\phi + c\phi s\theta s\psi) \\ u c\theta c\phi \\ -(f_2 - f_4)\ell s\theta + c\theta(\sum \tau_i c\phi + (f_3 - f_1)\ell s\phi) \\ (f_3 - f_1)\ell c\phi - \sum \tau_i s\phi \\ (f_2 - f_4)\ell \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} c\phi c\psi s\theta + s\phi s\psi & 0 & 0 & 0 \\ -c\psi s\phi + c\phi s\theta s\psi & 0 & 0 & 0 \\ c\theta c\phi & 0 & 0 & 0 \\ 0 & -s\theta & c\theta s\phi & c\theta c\phi \\ 0 & 0 & c\phi & -s\phi \\ 0 & 1 & 0 & 0 \end{bmatrix}}_{S^T} \begin{Bmatrix} T \\ \tau_x \\ \tau_y \\ \tau_z \end{Bmatrix}$$

$$\Rightarrow \quad \ddot{q} = M(q)^{-1}\left(S^T(q)\,u - g(q)\right)$$